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Python PetPal Simulator

An Object-Oriented Virtual Companion & Adventure Game
Course: Introduction to Python Programming (102040306) | Semester 3 (Odd)
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1. Abstract

The **Python PetPal Simulator** is an interactive console-based companion application implemented in pure standard Python. It demonstrates core software engineering paradigms including Object-Oriented Programming (OOP), state-driven mood resolution, procedural adventure encounters, experience-based leveling, and CSV-style persistent file storage in under 140 lines of code.

2. Objectives

- Encapsulate pet physiology (Hunger, Happiness, Energy, Level, XP) within an object-oriented class.
- Derive dynamic behavioral mood indicators using priority-based conditional logic.
- Facilitate turn-based user interactions with balanced state trade-offs.
- Maintain persistent game continuity via formatted text file serialization (pet_save.txt).
- Guarantee robust execution across platforms with structured exception handling.

3. System Architecture & Class Design

The system is centered around the **Pet** class, which acts as the primary data model and business logic container:

Component	Type	Description
name, species	Attributes (str)	Pet identity parameters defined during adoption.
hunger, happiness, energy	Attributes (int)	Vital statistics clamped within [0, 100].
level, xp	Attributes (int)	Progressive leveling metrics: XP threshold = Level * 40.
get_mood()	Method (str)	Priority evaluation: Tired -> Hungry -> Happy -> Neutral.
status()	Method	Draws ASCII progress bar dashboard with current mood.
feed()	Method	Pantry selection modifying hunger and granting XP.
adventure()	Method	Procedural encounter generator using random.choice().
sleep()	Method	Fully recharges energy to 100% while advancing hunger.
save() / load_pet()	Methods / Routines	Disk persistence using comma-separated text storage.

4. Syllabus & Practical Mapping

Syllabus Practical	Topic	Implementation Details in Project
Practical 2 & 3	Conditionals & Loops	Finite game loop, mood priority cascade, stat clamping.

Practical 5	Data Structures	MOOD_ART mapping dictionary, food menus, encounter tuples.
Practical 6	File I/O & Strings	State persistence to pet_save.txt using .split() and formatting.
Practical 7	Exception Handling	try...except blocks guarding I/O and input conversion errors.
Practical 8	OOP & Classes	Class definition, constructor, methods, state encapsulation.
Practical 9	Standard Library	random.choice() for stochastic RPG adventure encounters.

5. Conclusion

The Python PetPal Simulator demonstrates that comprehensive software design principles—such as OOP, data normalization, error handling, and file persistence—can be effectively demonstrated in an engaging, production-grade terminal application without unnecessary framework overhead.